Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R12-E1_000
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous)

Name	SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous)
Description	
Analysis	Stress, strain/ simplified loading

Design

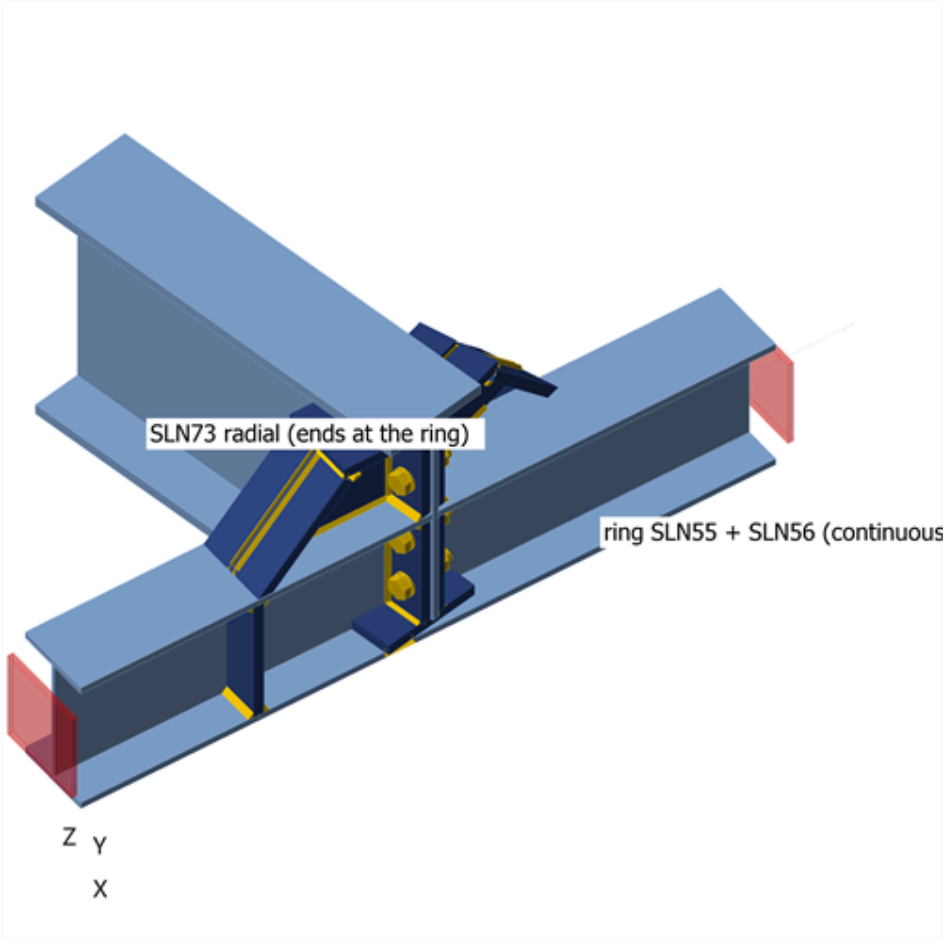
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN73 radial (ends at the ring)	1 - UB457x191x98	0.0	0.0	0.0	76	0	0
ring SLN55 + SLN56 (continuous)	2 - UB254x146x43	-100.0	0.0	0.0	0	0	-104

Supports and forces

Name	Support	Forces in	X [mm]
SLN73 radial (ends at the ring) / begin		Position	0
ring SLN55 + SLN56 (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
ring SLN55 + SLN56 (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0



SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous), Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M24 10.9	24	900.0	1000.0	452

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN73 radial (ends at the ring) / Begin	2.1	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_102	SLN73 radial (ends at the ring) / Begin	9.2	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_103	SLN73 radial (ends at the ring) / Begin	9.6	-0.1	-86.2	-0.1	-45.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_104	SLN73 radial (ends at the ring) / Begin	7.9	0.0	-86.4	-0.1	-45.4	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_105	SLN73 radial (ends at the ring) / Begin	8.8	-0.1	-86.3	-0.1	-45.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_106	SLN73 radial (ends at the ring) / Begin	8.0	0.0	-86.3	-0.1	-45.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_107	SLN73 radial (ends at the ring) / Begin	1.8	-0.1	-7.8	0.0	-4.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_108	SLN73 radial (ends at the ring) / Begin	1.7	0.1	-8.4	0.0	-4.6	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_109	SLN73 radial (ends at the ring) / Begin	1.4	0.1	-8.1	0.0	-4.5	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_110	SLN73 radial (ends at the ring) / Begin	0.6	-0.1	-8.1	0.0	-4.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_111	SLN73 radial (ends at the ring) / Begin	7.4	-0.1	-61.7	0.0	-32.6	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_112	SLN73 radial (ends at the ring) / Begin	6.3	0.1	-62.3	0.0	-32.8	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_113	SLN73 radial (ends at the ring) / Begin	6.7	0.0	-62.0	0.0	-32.7	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_114	SLN73 radial (ends at the ring) / Begin	5.6	-0.1	-62.0	0.0	-32.7	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_115	SLN73 radial (ends at the ring) / Begin	2.4	-0.1	-13.1	0.0	-7.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_116	SLN73 radial (ends at the ring) / Begin	2.1	0.1	-13.6	0.0	-7.4	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_117	SLN73 radial (ends at the ring) / Begin	2.0	0.0	-13.3	0.0	-7.3	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_118	SLN73 radial (ends at the ring) / Begin	1.1	-0.1	-13.3	0.0	-7.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_119	SLN73 radial (ends at the ring) / Begin	11.2	-0.1	-89.8	-0.1	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_120	SLN73 radial (ends at the ring) / Begin	8.1	0.0	-89.8	-0.1	-47.2	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_121	SLN73 radial (ends at the ring) / Begin	8.5	-0.1	-89.8	-0.1	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_122	SLN73 radial (ends at the ring) / Begin	7.6	0.0	-89.8	-0.1	-47.2	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_123	SLN73 radial (ends at the ring) / Begin	10.8	-0.1	-68.9	0.0	-36.3	0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_124	SLN73 radial (ends at the ring) / Begin	6.7	0.1	-68.9	0.0	-36.3	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_125	SLN73 radial (ends at the ring) / Begin	6.2	0.0	-68.9	0.0	-36.3	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_126	SLN73 radial (ends at the ring) / Begin	4.8	-0.1	-68.9	0.0	-36.3	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_127	SLN73 radial (ends at the ring) / Begin	5.7	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_128	SLN73 radial (ends at the ring) / Begin	2.5	0.1	-20.2	0.0	-10.9	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_129	SLN73 radial (ends at the ring) / Begin	1.5	0.0	-20.3	0.0	-10.9	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_130	SLN73 radial (ends at the ring) / Begin	0.3	-0.1	-20.3	0.0	-10.9	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_135	SLN73 radial (ends at the ring) / Begin	7.7	0.0	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_141	SLN73 radial (ends at the ring) / Begin	2.3	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_142	SLN73 radial (ends at the ring) / Begin	1.8	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_143	SLN73 radial (ends at the ring) / Begin	2.1	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_144	SLN73 radial (ends at the ring) / Begin	2.0	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_145	SLN73 radial (ends at the ring) / Begin	10.1	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_146	SLN73 radial (ends at the ring) / Begin	8.3	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_147	SLN73 radial (ends at the ring) / Begin	9.4	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_148	SLN73 radial (ends at the ring) / Begin	9.0	0.0	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_151	SLN73 radial (ends at the ring) / Begin	6.9	-0.1	-46.4	0.0	-22.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_152	SLN73 radial (ends at the ring) / Begin	9.3	0.0	-63.5	0.0	-35.9	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_153	SLN73 radial (ends at the ring) / Begin	9.2	-0.1	-89.4	0.0	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_154	SLN73 radial (ends at the ring) / Begin	7.0	-0.1	-46.8	0.0	-22.5	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	8.0 > 5.0%	**Not OK!**
Bolts	60.4 < 100%	OK
Welds	105.3 > 100%	**Not OK!**
Buckling	Not calculated	

Plates

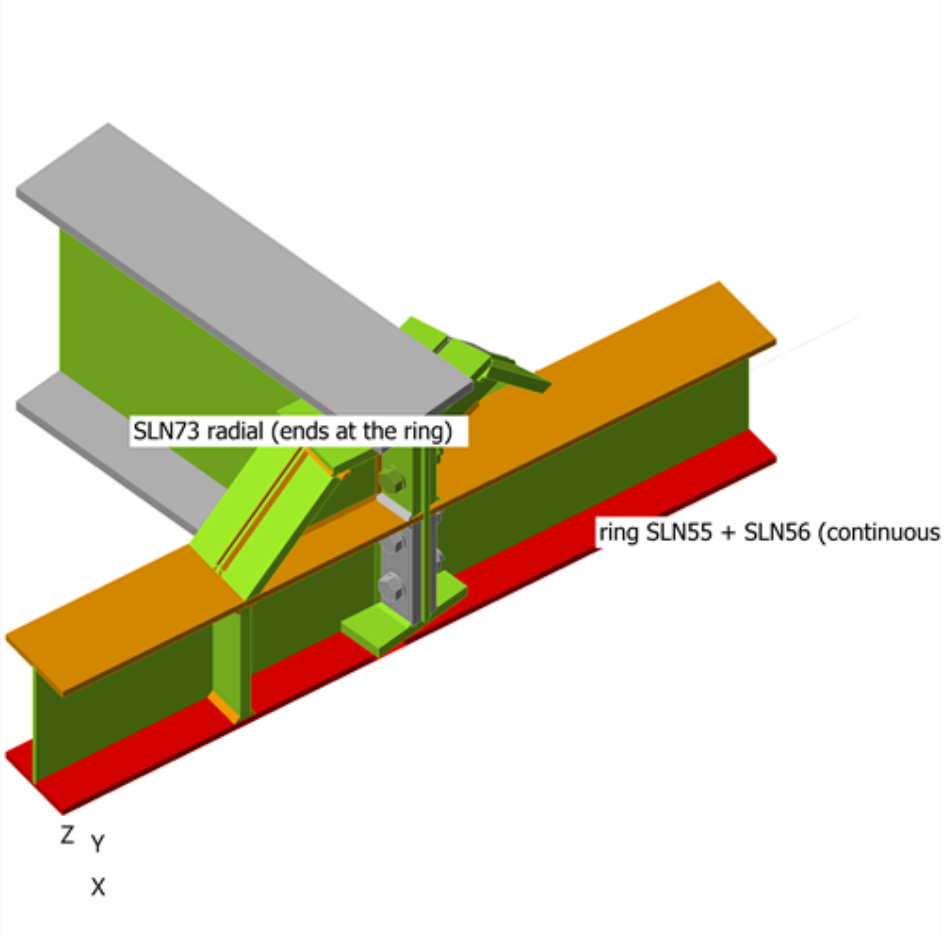
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN73 radial (ends at the ring)-bfl 1	19.6	C LC_121	14.1	0.0	0.0	OK
SLN73 radial (ends at the ring)-tfl 1	19.6	C LC_122	165.1	0.0	17.9	OK
SLN73 radial (ends at the ring)-w 1	11.4	C LC_122	273.0	0.0	68.1	OK
ring SLN55 + SLN56 (continuous)-bfl 1	12.7	C LC_119	291.9	8.0	0.0	**Not OK!**
ring SLN55 + SLN56 (continuous)-tfl 1	12.7	C LC_135	283.2	3.9	0.0	OK
ring SLN55 + SLN56 (continuous)-w 1	7.2	C LC_119	276.0	0.5	0.0	OK
395x145x20 fin plate upper (SLN55 side)	20.0	C LC_119	275.4	0.2	588.6	OK
395x145x20 fin plate lower outer (SLN55 side)	20.0	C LC_119	138.4	0.0	12.6	OK
395x145x20 fin plate lower inner (SLN55 side)	20.0	C LC_122	184.8	0.0	83.5	OK
395x145x20 fin plate upper (SLN56 side)	20.0	C LC_135	275.2	0.1	650.7	OK
395x145x20 fin plate lower outer (SLN56 side)	20.0	C LC_119	100.2	0.0	14.8	OK
395x145x20 fin plate lower inner (SLN56 side)	20.0	C LC_122	131.5	0.0	87.0	OK
20 ring stiffener at 315 (SLN55 side)	20.0	C LC_122	281.1	2.9	0.0	OK
10 bracket web plate (SLN55 side)	10.0	C LC_122	275.2	0.1	0.0	OK
20 bracket top plate a (SLN55 side)	20.0	C LC_122	275.1	0.0	86.3	OK
20 inclined stiffener a (SLN55 side)	20.0	C LC_122	215.4	0.0	0.0	OK
20 bracket top plate b (SLN55 side)	20.0	C LC_122	275.4	0.2	69.4	OK
20 inclined stiffener b (SLN55 side)	20.0	C LC_122	276.5	0.7	0.0	OK
20 ring stiffener at 315 (SLN56 side)	20.0	C LC_135	276.3	0.6	0.0	OK
10 bracket web plate (SLN56 side)	10.0	C LC_122	275.4	0.2	0.0	OK
20 bracket top plate a (SLN56 side)	20.0	C LC_135	277.2	1.1	0.0	OK
20 inclined stiffener a (SLN56 side)	20.0	C LC_122	275.4	0.2	0.0	OK
20 bracket top plate b (SLN56 side)	20.0	C LC_122	275.0	0.0	218.2	OK
20 inclined stiffener b (SLN56 side)	20.0	C LC_122	275.3	0.2	0.0	OK
185x70x20 plates under the fins (section E, 2 x 70 + web)	20.0	C LC_121	275.0	0.0	0.0	OK

Design data

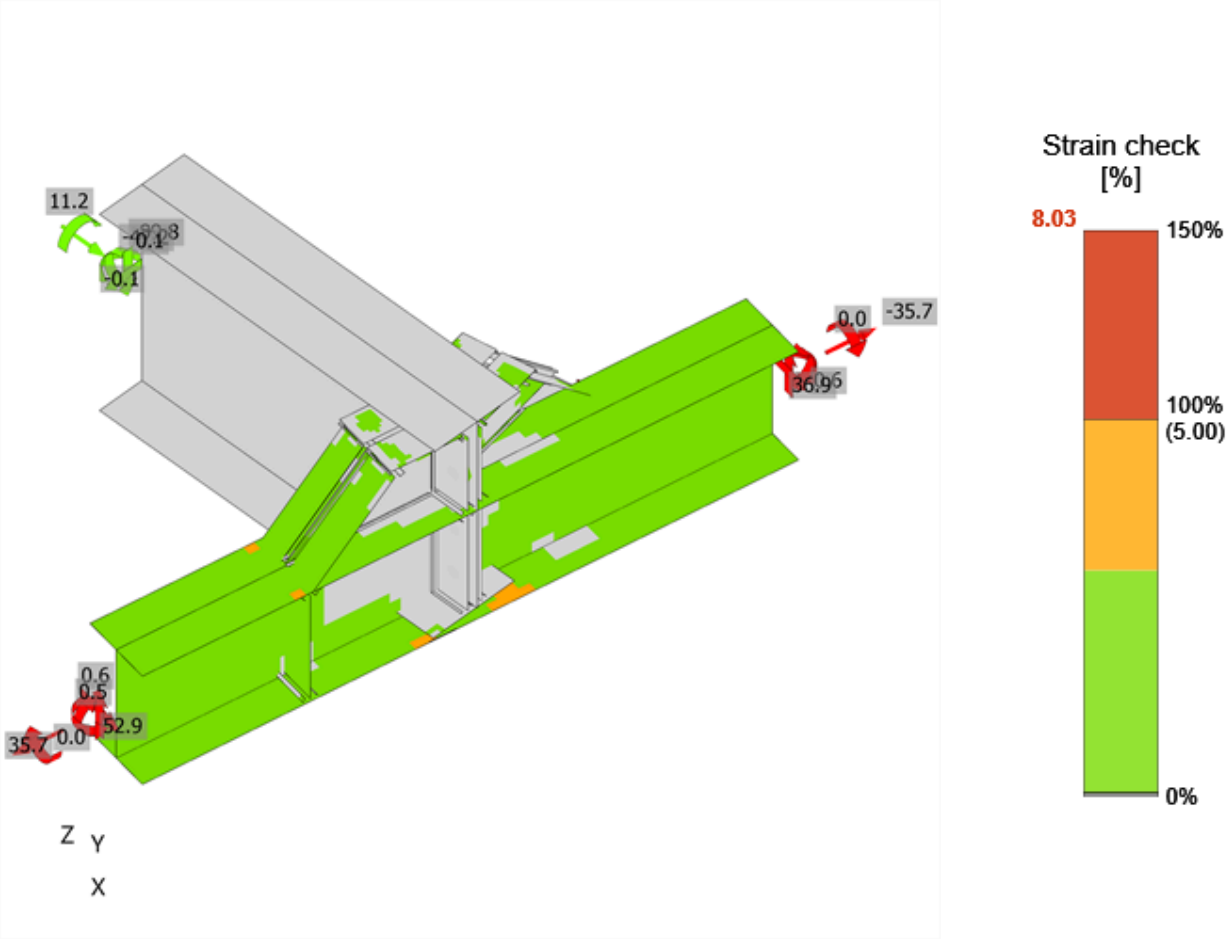
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

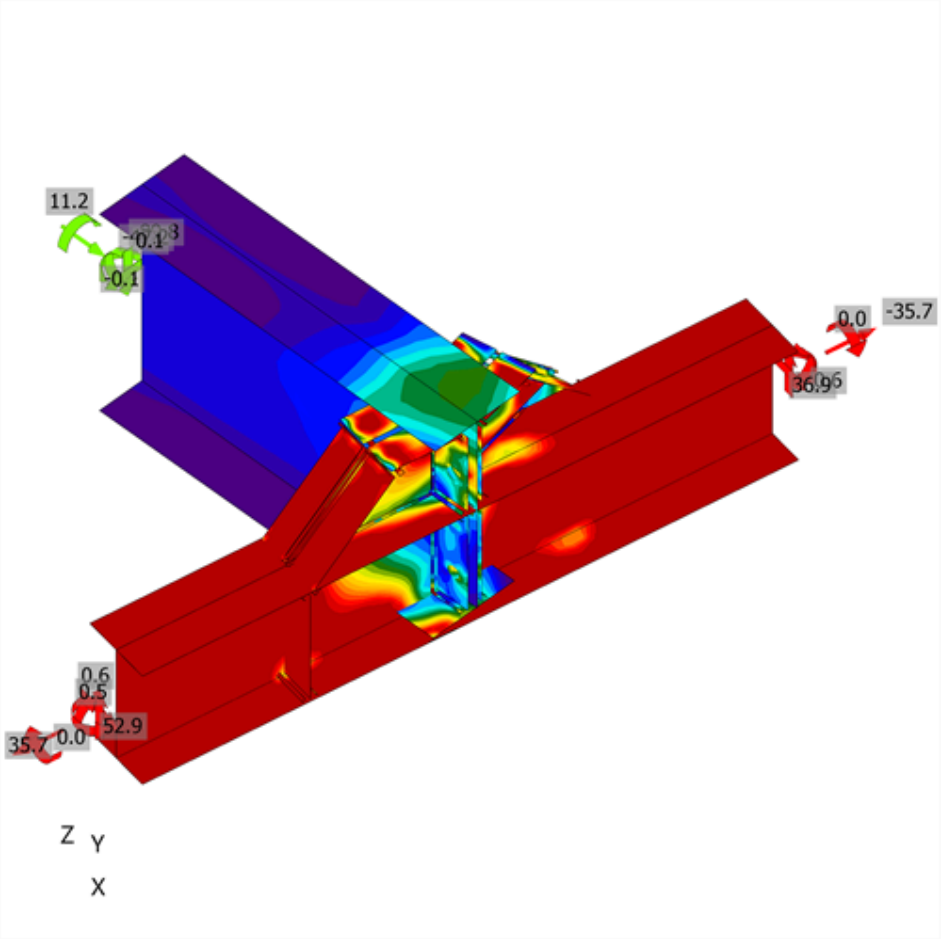
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

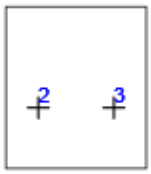
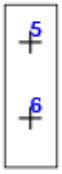
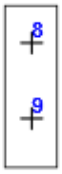


Strain check, C LC_119



Equivalent stress, C LC_119

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M24 10.9 - 1	C LC_122	35.7	37.9	176.5	14.0	40.6	36.9	OK	OK
	B3	M24 10.9 - 1	C LC_122	5.9	38.2	119.3	2.3	60.4	28.7	OK	OK
	B5	M24 10.9 - 1	C LC_122	6.3	12.3	235.3	2.5	9.7	10.5	OK	OK
	B6	M24 10.9 - 1	C LC_153	0.6	36.2	235.3	0.2	30.6	25.8	OK	OK
	B8	M24 10.9 - 1	C LC_119	0.2	13.3	235.3	0.1	10.5	9.5	OK	OK
	B9	M24 10.9 - 1	C LC_153	7.3	34.5	235.3	2.9	28.8	26.5	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M24 10.9 - 1	254.2	492.7	141.2

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
10 bracket web plate (SLN55 side)	20 bracket top plate a (SLN55 side)	-	128	-	-	-	-	-	-	-	-	OK	OK
		-	128	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN55 side)	20 bracket top plate b (SLN55 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN56 side)	20 bracket top plate a (SLN56 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN56 side)	20 bracket top plate b (SLN56 side)	-	128	-	-	-	-	-	-	-	-	OK	OK
		-	128	-	-	-	-	-	-	-	-	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower outer (SLN55 side)	◢ 4.2 ◣	213	C LC_119	278.9	0.0	-16.0	157.7	30.9	68.9	49.0	OK	OK
		◢ 4.2 ◣	213	C LC_119	396.7	0.1	91.6	12.3	222.5	98.0	61.6	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower inner (SLN55 side)	◢ 7.0 ◣	69	C LC_122	155.0	0.0	27.5	37.3	-79.8	40.0	31.0	OK	OK
		◢ 7.0 ◣	69	C LC_122	230.2	0.0	-56.8	-35.6	-123.8	56.9	36.1	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower inner (SLN55 side)	◢ 4.2 ◣	213	C LC_121	380.9	0.0	3.9	-191.9	-107.4	94.1	51.5	OK	OK
		◢ 4.2 ◣	213	C LC_119	272.4	0.0	41.7	-118.5	-100.6	67.3	34.3	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate upper (SLN56 side)	◢ 7.0 ◣	144	C LC_135	123.8	0.0	-56.6	-36.3	-52.2	30.6	23.6	OK	OK
		◢ 7.0 ◣	144	C LC_119	219.4	0.0	135.1	-96.3	-26.2	54.2	39.0	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower outer (SLN56 side)	◢ 7.0 ◣	69	C LC_120	109.2	0.0	61.4	47.4	21.7	27.0	20.7	OK	OK
		◢ 7.0 ◣	69	C LC_119	114.3	0.0	45.0	-56.7	-21.6	28.3	22.2	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower outer (SLN56 side)	◢ 4.2 ◣	213	C LC_153	343.3	0.0	-91.7	131.0	139.0	84.8	53.6	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	213	C LC_153	185.6	0.0	133.5	28.6	68.8	45.9	29.9	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower inner (SLN56 side)	◢ 7.0 ◣	69	C LC_122	97.5	0.0	-91.5	-13.9	13.4	29.6	19.7	OK	OK
		◢ 7.0 ◣	69	C LC_119	140.2	0.0	-5.4	78.6	-19.0	34.7	25.8	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower inner (SLN56 side)	◢ 4.2 ◣	213	C LC_153	330.1	0.0	-134.1	12.0	-173.8	81.6	42.8	OK	OK
		◢ 4.2 ◣	213	C LC_119	265.9	0.0	51.5	146.9	-33.1	65.7	35.1	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	20 ring stiffener at 315 (SLN56 side)	◢ 7.0 ◣	147	C LC_103	396.7	0.0	113.4	158.3	152.0	98.0	78.2	OK	OK
		◢ 7.0 ◣	146	C LC_122	308.4	0.0	-120.1	61.7	151.9	76.2	57.1	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 ring stiffener at 315 (SLN56 side)	◢ 7.0 ◣	146	C LC_153	396.7	0.1	122.8	-214.0	40.4	98.0	85.0	OK	OK
		◢ 7.0 ◣	146	C LC_153	403.3	4.1	-197.7	194.7	-57.2	99.7	99.1	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	10 bracket web plate (SLN56 side)	◢ 4.2 ◣	288	C LC_119	397.2	0.4	57.7	-102.4	202.5	98.1	66.4	OK	OK
		◢ 4.2 ◣	287	C LC_119	397.0	0.2	-149.2	-75.2	198.6	98.1	55.8	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener a (SLN56 side)	◢ 7.0 ◣	67	C LC_153	400.3	2.3	-123.1	183.0	-122.0	98.9	91.0	OK	OK
		◢ 7.0 ◣	67	C LC_122	400.9	2.6	-126.4	-201.8	86.6	99.0	81.8	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener b (SLN56 side)	◢ 7.0 ◣	67	C LC_153	400.1	2.2	129.2	-185.5	-115.7	98.9	91.9	OK	OK
		◢ 7.0 ◣	67	C LC_122	400.1	2.2	129.7	198.3	91.7	98.9	89.6	OK	OK
395x145x20 fin plate upper (SLN55 side)	10 bracket web plate (SLN55 side)	◢ 4.2 ◣	167	C LC_135	396.7	0.1	-230.4	-186.3	-6.8	98.0	69.3	OK	OK
		◢ 4.2 ◣	166	C LC_122	371.4	0.0	157.1	-178.0	78.0	91.8	71.3	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
395x145x20 fin plate upper (SLN56 side)	10 bracket web plate (SLN56 side)	◢ 4.2 ◣	167	C LC_121	402.2	3.4	145.5	-189.1	-105.3	99.4	84.6	OK	OK
		◢ 4.2 ◣	167	C LC_119	399.8	1.9	-303.8	-113.5	-98.1	98.8	81.3	OK	OK
20 bracket top plate a (SLN55 side)	20 inclined stiffener a (SLN55 side)	◢ 7.0 ◣	67	C LC_122	181.8	0.0	-177.4	23.0	0.7	57.3	28.7	OK	OK
		◢ 7.0 ◣	67	C LC_122	235.3	0.0	162.0	-97.7	-12.8	58.1	36.3	OK	OK
20 bracket top plate b (SLN55 side)	20 inclined stiffener b (SLN55 side)	◢ 7.0 ◣	67	C LC_122	215.4	0.0	-176.1	-71.1	-8.5	56.9	43.0	OK	OK
		◢ 7.0 ◣	67	C LC_122	276.4	0.0	-189.8	115.5	10.7	68.3	49.2	OK	OK
20 bracket top plate a (SLN56 side)	20 inclined stiffener a (SLN56 side)	◢ 7.0 ◣	67	C LC_122	397.3	0.4	9.6	-40.7	-225.7	98.2	77.4	OK	OK
		◢ 7.0 ◣	67	C LC_122	396.7	0.1	-79.4	45.5	-219.7	98.0	68.9	OK	OK
20 bracket top plate b (SLN56 side)	20 inclined stiffener b (SLN56 side)	◢ 7.0 ◣	67	C LC_122	397.7	0.6	-39.4	6.7	-228.4	98.3	80.8	OK	OK
		◢ 7.0 ◣	67	C LC_122	396.7	0.1	-29.2	-3.5	-228.4	98.0	70.3	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate upper (SLN55 side)	◢ 7.0 ◣	141	C LC_122	341.0	0.0	-148.7	-90.0	-152.6	84.3	54.2	OK	OK
		◢ 7.0 ◣	141	C LC_122	289.8	0.0	-43.2	-59.6	154.4	71.6	38.9	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower outer (SLN55 side)	◢ 7.0 ◣	65	C LC_122	184.3	0.0	-61.6	-59.4	80.7	45.5	28.5	OK	OK
		◢ 7.0 ◣	65	C LC_122	122.8	0.0	76.1	-0.1	55.6	30.3	24.0	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	20 ring stiffener at 315 (SLN55 side)	◢ 7.0 ◣	37	C LC_120	412.6	9.8	-122.0	-208.6	-90.8	101.9	100.7	OK	**Not OK!**
		◢ 7.0 ◣	37	C LC_122	403.4	4.2	277.1	-106.8	-131.3	99.7	96.4	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 ring stiffener at 315 (SLN55 side)	◢ 7.0 ◣	37	C LC_120	426.0	18.1	-87.9	-215.8	-106.4	105.3	101.7	OK	**Not OK!**
		◢ 7.0 ◣	37	C LC_120	424.5	17.2	-299.0	164.8	-55.8	104.9	101.6	OK	**Not OK!**

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
ring SLN55 + SLN56 (continuous)-tfl 1	10 bracket web plate (SLN55 side)	◢ 4.2 ◣	189	C LC_122	396.7	0.0	-288.1	-89.4	-129.6	98.0	59.5	OK	OK
		◢ 4.2 ◣	189	C LC_122	396.7	0.1	-47.4	225.9	-26.0	98.0	62.4	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener b (SLN55 side)	◢ 7.0 ◣	39	C LC_122	410.1	8.3	277.0	-162.3	64.3	101.3	100.0	OK	**Not OK!**
		◢ 7.0 ◣	39	C LC_122	401.4	2.9	-54.0	-229.5	8.1	99.2	92.9	OK	OK
185x70x20 plates under the fins (section E, 2 x 70 + web)	ring SLN55 + SLN56 (continuous)-bfl 1	◢ 7.0	82	C LC_119	424.4	17.2	-120.2	-107.9	-208.8	104.9	100.0	OK	**Not OK!**
185x70x20 plates under the fins (section E, 2 x 70 + web)	ring SLN55 + SLN56 (continuous)-bfl 1	◢ 7.0	50	C LC_119	406.2	5.9	124.4	98.6	200.3	100.4	99.0	OK	**Not OK!**
185x70x20 plates under the fins (section E, 2 x 70 + web)	395x145x20 fin plate lower outer (SLN55 side)	◢ 4.2 ◣	66	C LC_119	396.7	0.0	89.5	36.1	-220.2	98.0	64.3	OK	OK
		◢ 4.2 ◣	66	C LC_119	324.1	0.0	41.9	185.4	-6.8	80.1	52.5	OK	OK
185x70x20 plates under the fins (section E, 2 x 70 + web)	395x145x20 fin plate lower inner (SLN55 side)	◢ 4.2 ◣	66	C LC_121	219.3	0.0	61.9	-121.2	8.5	54.2	38.7	OK	OK
		◢ 4.2 ◣	66	C LC_119	396.7	0.1	21.8	-156.3	-167.0	98.0	64.9	OK	OK
185x70x20 plates under the fins (section E, 2 x 70 + web)	395x145x20 fin plate lower outer (SLN56 side)	◢ 4.2 ◣	66	C LC_153	391.1	0.0	-17.1	158.1	-160.9	96.6	62.4	OK	OK
		◢ 4.2 ◣	66	C LC_119	183.4	0.0	-106.9	-85.9	4.9	45.3	35.6	OK	OK
185x70x20 plates under the fins (section E, 2 x 70 + web)	395x145x20 fin plate lower inner (SLN56 side)	◢ 4.2 ◣	66	C LC_153	313.7	0.0	-44.7	179.2	5.1	77.5	42.1	OK	OK
		◢ 4.2 ◣	66	C LC_153	375.6	0.0	-79.3	55.3	204.6	92.8	59.2	OK	OK
10 bracket web plate (SLN55 side)	20 inclined stiffener a (SLN55 side)	◢ 4.2 ◣	222	C LC_122	241.3	0.0	-31.8	62.7	123.0	59.6	43.3	OK	OK
		◢ 4.2 ◣	222	C LC_122	187.6	0.0	98.4	-3.9	92.2	46.4	33.8	OK	OK
10 bracket web plate (SLN55 side)	20 inclined stiffener b (SLN55 side)	◢ 4.2 ◣	222	C LC_122	304.3	0.0	-229.5	-64.2	-95.9	75.2	44.5	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	222	C LC_122	300.1	0.0	138.6	-25.3	151.6	74.2	46.0	OK	OK
10 bracket web plate (SLN56 side)	20 inclined stiffener a (SLN56 side)	◢ 4.2 ◣	222	C LC_122	397.6	0.6	125.3	89.2	198.8	98.3	89.7	OK	OK
		◢ 4.2 ◣	222	C LC_119	397.4	0.5	147.7	-193.9	-88.2	98.2	73.8	OK	OK
10 bracket web plate (SLN56 side)	20 inclined stiffener b (SLN56 side)	◢ 4.2 ◣	222	C LC_119	397.4	0.5	-295.6	-121.0	94.2	98.2	86.4	OK	OK
		◢ 4.2 ◣	222	C LC_122	397.7	0.6	-59.5	202.4	-102.7	98.3	79.5	OK	OK
395x145x20 fin plate upper (SLN55 side)	20 bracket top plate a (SLN55 side)	◢ 4.2 ◣	65	C LC_122	396.9	0.2	-159.4	-209.6	-10.9	98.1	74.7	OK	OK
		◢ 4.2 ◣	65	C LC_122	396.9	0.2	-146.7	-201.2	-69.5	98.1	69.1	OK	OK
395x145x20 fin plate upper (SLN55 side)	20 bracket top plate b (SLN55 side)	◢ 4.2 ◣	68	C LC_122	399.5	1.8	-163.9	210.0	12.8	98.7	92.0	OK	OK
		◢ 4.2 ◣	68	C LC_122	399.3	1.7	-17.4	225.9	-45.0	98.7	88.5	OK	OK
395x145x20 fin plate upper (SLN56 side)	20 bracket top plate a (SLN56 side)	◢ 4.2 ◣	68	C LC_135	399.0	1.5	-256.5	176.0	12.6	98.6	90.5	OK	OK
		◢ 4.2 ◣	68	C LC_135	398.7	1.3	-198.7	196.7	33.5	98.5	90.5	OK	OK
395x145x20 fin plate upper (SLN56 side)	20 bracket top plate b (SLN56 side)	◢ 4.2 ◣	65	C LC_122	397.2	0.4	-166.4	-204.7	38.1	98.1	78.0	OK	OK
		◢ 4.2 ◣	65	C LC_122	397.3	0.4	184.7	-187.0	79.3	98.2	82.8	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5